Thesis Title Goes Here

Firstname Lastname

Munich 20XX

Thesis Title Goes Here

Firstname Lastname

Dissertation
an der Fakultät für Physik
der Ludwig–Maximilians–Universit"at
München

vorgelegt von
Firstname Lastname
aus City, Country

München, den DD. Month YYYY

Erstgutachter: Prof. Dr. First Reviewer

Zweitgutachter: Prof. Dr. Second Reviewer

Tag der mündlichen Prüfung: DD. Month YYYY

Contents

Abstract **xi**

1 Introduction **1**

1.1 Background and Motivation 1

1.2 Research Questions 1

1.3 Structure of This Thesis 1

2 Data, Observations, and Methods **3**

2.1 Sample and Survey Overview 3

2.1.1 Selection Criteria 3

2.2 Analysis Method 4

2.3 Summary 4

A Supplementary Material **5**

B Acronyms and Abbreviations **7**

Acknowledgments **10**

List of Figures

2.1 Short caption for list of figures 4

List of Tables

2.1 Sample summary 4

Abstract

Write your abstract here. Aim for a maximum of one page summarising the motivation, methods, key results, and conclusions of the thesis.

CHAPTER 1

Introduction

1.1 Background and Motivation

Provide the scientific context and motivation for the thesis here. This section should introduce the key concepts and explain why the research problem is important (J. Smith & A. B. Jones, 2020).

A citation using C. Brown et al. (2021) integrates the author name into the sentence. For parenthetical references use (J. Smith & A. B. Jones, 2020). Prefix notes work as follows: (e.g., J. Smith & A. B. Jones, 2020).

1.2 Research Questions

State the specific questions addressed in this thesis.

1.3 Structure of This Thesis

Chapter 2 presents ... Chapter 3 describes ...

CHAPTER 2

Data, Observations, and Methods

This chapter demonstrates key LaTeX formatting features available in this template: running text, mathematical expressions, footnotes, a table, and a figure placeholder.

2.1 Sample and Survey Overview

Describe your dataset here (C. Brown et al., 2021). Inline citations use J. Smith & A. B. Jones (2020); parenthetical citations use (e.g., J. Smith & A. B. Jones, 2020).[1]

A summary of the sample properties is given in Table 2.1. All results assume a flat ΛCDM cosmology with $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\mathrm{m} = 0.3$.

2.1.1 Selection Criteria

Objects were selected according to the following criterion:

$$\log\left(\frac{L}{L_\odot}\right) > 10, \tag{2.1}$$

yielding a final sample of N sources spanning a redshift range $z_\mathrm{min} < z < z_\mathrm{max}$.

[1] Footnotes look like this. Keep them concise.

2.2 Analysis Method

Each source was modelled using a Sérsic profile (J. Smith & A. B. Jones, 2020),

$$I(r) = I_e \exp\left\{-b_n\left[\left(\frac{r}{r_e}\right)^{1/n} - 1\right]\right\}, \tag{2.2}$$

where r_e is the half-light radius, I_e is the surface brightness at r_e, and n is the Sérsic index.

Figure 2.1 shows an illustrative example.

Figure placeholder — replace with `\includegraphics{your_figure.pdf}`

Figure 2.1. Full figure caption goes here. Describe all panels. Left: … Right: …

Table 2.1. Summary of the sample used in this work.

Property	Value	Unit
Total sources	N	—
Redshift range	z_1–z_2	—
Median redshift	$\bar{z}$	—
Median $\log L$	$X.X$	$L_\odot$

2.3 Summary

1. The sample contains N sources selected via Equation (2.1).
2. Surface-brightness profiles were fitted with the Sérsic model (Equation 2.2).
3. Results are presented in Chapter 3.

APPENDIX A

Supplementary Material

Include additional tables, figures, or derivations here.

APPENDIX B

Acronyms and Abbreviations

AGN	Active Galactic Nucleus
SMBH	Supermassive Black Hole
…	…

Bibliography

Brown, C., White, D. E., & Green, F. 2021, Another Example Article, MNRAS, 500, 100, doi:10.1093/mnras/example

Smith, J., & Jones, A. B. 2020, An Example Article Title, ApJ, 900, 1, doi:10.3847/1538-4357/example

Acknowledgments

Write your acknowledgments here.

Curriculum Vitae

Firstname Lastname

Date of Birth	DD Month YYYY, City, Country
Education	BSc in Subject, University (YYYY)
	MSc in Subject, University (YYYY)
	PhD candidate, LMU München (YYYY–)
Publications	List key publications here